

Selecting Equipment and Technology Scientists collect and analyze data using tools that range from simple meter sticks to complex hardware that measures leaf nitrogen content. Data are often gathered directly by hardware controlled by computers running software that organizes and analyzes results. Statistical analysis helps determine if an experimental treatment is significantly different from controls.

Sources of Error Researchers must avoid errors in data collection and analysis. Tools used to measure the size and mass of marsh grasses, for example, have limited accuracy. Data analysis and sample size must be chosen carefully. The larger the sample size, the more reliably researchers can analyze variation within each group, and evaluate differences between experimentals and controls.

Interpreting Data and Drawing Conclusions Data analysis may lead to conclusions that support or refute the hypothesis being tested. Often, new data indicate that a hypothesis is on the right track, but is off-base about a few details. New questions lead to new and revised hypotheses, which are tested with new experiments that involve better control of variables or other changes in experimental design.

When Experiments Aren't Possible Not all hypotheses can be tested by experiments. Animal behavior researchers, for example, might propose hypotheses about how groups of animals interact in nature. These hypotheses are tested by field observations designed to disturb natural behavior as little as possible. Analysis of data from these observations may lead to new hypotheses that can be tested in different ways. If investigations suggest, for example, that members of a group are related to one another, genetic tests can gather data that support or reject that hypothesis.

Sometimes ethics prevent certain types of experiments. For example, in medical research, when a chemical is suspected as a cause of cancer, researchers do not purposely expose volunteers to the chemical. They study people who have already been exposed, using those who have not as the control group.

 **READING CHECK Describe** What is the difference between quantitative data and qualitative data?

Analyzing Data

What's in a Diet?

The circle graph shows the diet of the siamang gibbon, a type of ape found in the rain forests of Southeast Asia.

1. **Analyze Graphs** According to the circle graph, which plant parts do siamangs rely on most as a food source?
2. **Form a Hypothesis** How would siamangs be affected if the rain forests they live in produced less fruit? Explain your reasoning.

